Nucleo-WL55JC

The Nucleo-WL55JC is a development board for the STM32WL55 SoC from ST. It features 64 I/O, 3 onboard LEDs and buttons, integrated stlink for easy debug and flashing and on-board LoRa receiver with external antenna. NSH can be easily access via virtual serial port from usb.

Features

- STM32WL55JC MCU, 256K FLASH, 64K SRAM
- 32768 Hz LSE crystal
- 32 MHz HSE crystal
- Embedded stlink-v3 debugger (debug/flash and virtual serial port)
- Reset button
- 3 user programmable LEDs
- 3 user programmable buttons
- Power indicator LED
- LoRa radio with antenna
- 64 Nucleo I/O
- Arduino compatible pinout

Pin Mapping

Pin mapping can be altered by (de)soldering bridges, by default board uses following mapping:

Pin	Signal	Notes
PA2	LPUART1 TX	Virtual serial port over usb
PA3	LPUART2 RX	Virtual serial port over usb
PB6	USART1 TX	D1 on Arduino pinout
PB7	USART1 RX	R0 on Arduino pinout
PA0	Button 1	
PA1	Button 2	
PC6	Button 3	
PB15	Blue LED	Active HIGH
PB11	Red LED	Active HIGH

Pin	Signal	Notes
PB9	Green LED	Active HIGH

Default Peripherals Configuration

LED

Green and Red LEDs are used by the system at boot to show system state. Once system is booted these LEDs are for user to control. When CONFIG_ARCH_LEDS is set, Blue LED is reserved by OS for reporting system status. When CONFIG_ARCH_LEDS is not set, OS state won't be reported on any of the LEDs and all 3 LEDs are available for user right from the start.

Serial Console

There are 2 serial ports - USART1 and LPUART1.

USART1 is connected to arduino D0/D1 pin and LPUART is connected to stlink that provides virtual serial port.

NSH is configured to use LPUART and virtual serial port. After flashing you can open /dev/ttyACM0 (may change depending on your system) and nsh prompt will be waiting for you there. Serial device does not disappear when flashing and resetting board - it can be left opened and flashing will work without problems.

Configurations

nsh

Basic NuttShell configuration (console enabled in LPUART1, exposed via USB connection, at 115200 bps 8n1).

demo

This configuration contains demonstration code and programs to show what board is capable of. These programs and settings are usually absolutely useless in production code and are here just to show, by example, what you can do with the board.

Flash & Debug

Both flashing and debugging are done using the embedded stlink-v3 debugger. OpenOCD can be invoked in the following way to flash:

```
openocd -f interface/stlink.cfg -f target/stm32wlx.cfg \
    -c "program nuttx.bin exit 0x08000000"
```

To debug attach openocd to stlink with command:

```
openocd -f interface/stlink.cfg -f target/stm32wlx.cfg
```

start gdb:

```
arm-none-eabi-gdb ./nuttx --tui
```

attach to gdb server:

```
target remote localhost:3333
```

(optionally) reset program and start from the very beginning:

```
monitor reset halt
```

Remember to generate debug symbol by setting CONFIG_DEBUG_SYMBOLS and optionally (for more natural flow in gdb step) suppress optimization by setting CONFIG_DEBUG_NOOPT.